

Multi-Kernelized Fuzzy Granular Outlier Detector

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I. INTRODUCTION

Abstract— Fuzzy information granulation theory offers an effective framework for identifying uncertain outliers. However, existing kernelized fuzzy outlier detectors predominantly rely on a single kernel function, overlooking the potential of integrating diverse kernel information, which may lead to biases in outlier detection. Multi-kernel information fusion, by leveraging multiple kernel functions to extract information and employing information fusion strategies, can more comprehensively capture the characteristics of data distributions. Building on this, this study proposes a Multi-Kernelized Fuzzy Granular Outlier Detector (MKFGOD). The proposed detector first employs multiple kernel functions to extract fuzzy relation matrices from the data. It then integrates the multi-kernel information using various fusion strategies. Based on the fused multi-kernelized fuzzy relation matrix, the detector defines multi-kernelized fuzzy k-nearest neighbors and subsequently calculates the multi-kernelized fuzzy neighborhood density. Finally, the outlier degree of each sample is quantified by calculating the deviation in the multi-kernelized fuzzy neighborhood density. Experimental validation on multiple public datasets demonstrates that the proposed method significantly outperforms existing single-kernel and mainstream outlier detectors in terms of detection performance. This study not only enhances the effectiveness of outlier detection but also provides novel theoretical and practical guidance for the application of multi-kernel information fusion in fuzzy information granulation. Furthermore, it underscores the critical role of information fusion in complex data analysis.

Index Terms—Fuzzy information granulation theory, multi-kernel information fusion, kernelized fuzzy relations, fuzzy k-nearest neighbor, outlier detection.

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The code is publicly available online at <https://github.com/BELLoney/MKFGOD>.

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OUTLIER detection is a critical task in data mining, aimed at identifying samples that significantly deviate from the normal population or established patterns. In real-world applications, labeled data is often scarce, making unsupervised outlier detection a focal point of extensive research and application. It has been widely adopted in domains such as network intrusion detection [1] and financial fraud identification [2]. Fuzzy Information Granulation (FIG) theory leverages fuzzy relations to decompose complex data into fuzzy information granules, capturing data features in a way that closely mirrors human cognitive processes. Data uncertainty is a key characteristic of outliers, and FIG's ability to effectively handle such uncertainty makes it an ideal tool for outlier detection [3], [4], [5], [6]. Existing outlier detection methods encompass traditional methods, such as statistics-based, distance-based, and density-based methods, as well as recent advancements in deep learning, ensemble learning techniques, and granular-ball computing [7], [8].

Statistics-based methods detect outliers by analyzing statistical metrics, typically assuming data follows a specific probability distribution. However, real-world data often exhibits complex or unknown distributions, leading to poor detection performance. Distance-based methods rely on various distance metrics, identifying samples far from the majority as outliers, but in high-dimensional data, distance measures become unreliable as inter-sample distances converge, making outliers hard to distinguish [7], [9]. Similarly, density-based methods flag samples in low-density regions as outliers, yet density estimation falters in high-dimensional spaces, compromising performance [7], [9]. Recently, deep learning-based methods [10], [11] have gained attention by constructing neural networks to learn complex data patterns and features, distinguishing inliers from outliers. However, these models typically require extensive training data to effectively learn robust representations of the normal data distribution. In many real-world applications where the dataset size is limited, deep learning models may struggle to capture the underlying manifold of normal data accurately, potentially leading to suboptimal detection performance. Ensemble learning-based methods [12], [13] combine multiple weak learners to create robust detection models, leveraging diversity and robustness to identify outliers, but they suffer from high complexity and time-consuming training and tuning processes. Additionally, Granular-Ball Computing (GBC) has recently emerged as an efficient strategy for data characterization and information granulation. By utilizing hyper-spheres of varying sizes to represent data distributions, GBC effectively strikes a balance between representation accuracy and computational efficiency.

Following its successful application in clustering tasks, such as density peaks clustering [14], manifold clustering [15], and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [16], GBC has been increasingly integrated into the field of outlier detection. For instance, Su et al. [4] pioneered the integration of GBC with FIG by developing a granular-ball fuzzy rough set model, where granular-ball fuzzy approximation accuracy was utilized as a key metric for outlier detection. Building upon this foundation, Su et al. [17] further unified GBC, FIG, and the local density estimation concept from the classic LOF algorithm into a cohesive framework, identifying outliers through a novel metric termed local granular-ball density. In contrast, FIG-based methods [4], [17], [18] effectively handle data uncertainty by decomposing datasets into fuzzy information granules and integrating multiple uncertainty metrics to detect outliers. These methods, also known as fuzzy granular outlier detection methods, have been increasingly explored in recent research due to their superior capability in managing uncertainty inherent in real-world data. To illustrate the effectiveness of FIG compared to traditional approaches, consider a sample on the boundary of a normal cluster. Traditional methods often use rigid thresholds, risking false positives for such boundary points. In contrast, FIG-based methods assign continuous membership degrees rather than binary labels. This preserves uncertainty information, enabling a more robust distinction between legitimate boundary samples and true outliers.

The core principle of FIG theory involves aggregating data elements into granules based on fuzzy relations, where each element's membership to a granule is represented by a continuous value between 0 and 1, rather than binary affiliation. FIG has been effectively applied to outlier detection tasks. For instance, Yuan et al. [18] introduced FIG to detect outliers in mixed-attribute data, proposing a method based on fuzzy rough granules. Su et al. [4] combined FIG with granular-ball computing to develop a granular-ball fuzzy rough set model, utilizing granular-ball fuzzy approximation accuracy for outlier detection. Yuan et al. [19] proposed a fuzzy rough density-based method grounded in FIG. Additionally, Wu et al. [20] integrated kernel functions with FIG, employing kernelized fuzzy relations to granulate data and detected outliers through kernelized fuzzy approximation accuracy.

Kernel functions are a cornerstone of machine learning, enabling the transformation of non-linearly separable data from its original space into a higher-dimensional feature space via non-linear mappings, thus achieving linear separability. By performing computations in the lower-dimensional space, kernel functions efficiently calculate inner products in the high-dimensional space indirectly. Kernel functions effectively tackle the challenges of classification in low-dimensional spaces and mitigate the computational burden associated with high-dimensional data processing. The method proposed by Wu et al. [20], which integrates kernel functions with FIG, has demonstrated promising results in outlier detection. However, as a single-kernel fuzzy granular outlier detection method, its reliance on a fixed and single-kernel mapping restricts its adaptability to diverse and complex data distributions, potentially leading to the loss of critical features during granulation and reducing detection

performance. Furthermore, this method lacks the benefits of multi-kernel information fusion, which could enhance the modeling of fuzzy relations by integrating diverse kernel mappings to capture complex inter-sample relationships.

To address the limitations of the above single-kernel fuzzy granular outlier detection method, which often struggles with diverse and complex data distributions due to reliance on a single-kernel mapping, information fusion provides a powerful technique. Information fusion integrates multiple data sources, feature representations, or models to achieve more robust and effective decision-making, effectively enhancing pattern recognition [3], [21]. Among its techniques, multi-kernel information fusion is particularly promising, as it combines diverse kernel functions to capture a wide range of data characteristics, thereby overcoming the constraints of single-kernel methods and improving the performance of outlier detection in complex datasets [22], [23], [24]. Based on these observations, our main hypothesis is that integrating multiple kernel functions through multi-kernel fusion can overcome the inductive bias of a fixed single-kernel mapping, thereby capturing diverse data characteristics more comprehensively and improving outlier detection performance.

Building on the above discussion, we propose the Multi-Kernelized Fuzzy Granular Outlier Detector (MKFGOD). The framework of MKFGOD is presented in Fig. 1. Specifically, this study focuses on detecting point-based outliers in multi-variate tabular data. While existing single-kernel fuzzy granular methods have shown promise, they face a critical bottleneck: the reliance on a fixed, single-kernel mapping often fails to capture the complex, non-linear distributions inherent in diverse datasets. To address this limitation, we propose a multi-kernel information fusion model that fuses multiple kernelized fuzzy relations to better capture complex inter-sample relationships, enhancing the modeling of traditional fuzzy relations and the handling of uncertainty information in data. Based on the derived multi-kernelized fuzzy information granules, we develop a novel density estimation method termed multi-kernelized fuzzy neighborhood density. We utilize its deviation to calculate outlier scores, which reflect the outlier degrees of each sample. Finally, outliers are detected by comparing each sample's outlier score against a predefined threshold. In summary, the contributions of this study are as follows.

- 1) We propose a multi-kernel information fusion model incorporating various fusion strategies to comprehensively capture the characteristics of data distributions.
- 2) Leveraging the fused multi-kernelized fuzzy relation, we introduce a density metric termed multi-kernelized fuzzy neighborhood density based on fuzzy k -nearest neighbors.
- 3) By integrating FIG theory with multi-kernel information fusion, we construct the Multi-Kernelized Fuzzy Granular Outlier Detector (MKFGOD).
- 4) Extensive experiments validate the effectiveness of MKFGOD.

The remaining structure of this study is organized as follows. Section II reviews traditional outlier detection methods and fuzzy granular outlier detection methods. Section III revisits the FIG theory, kernel functions, and a density-based anomaly

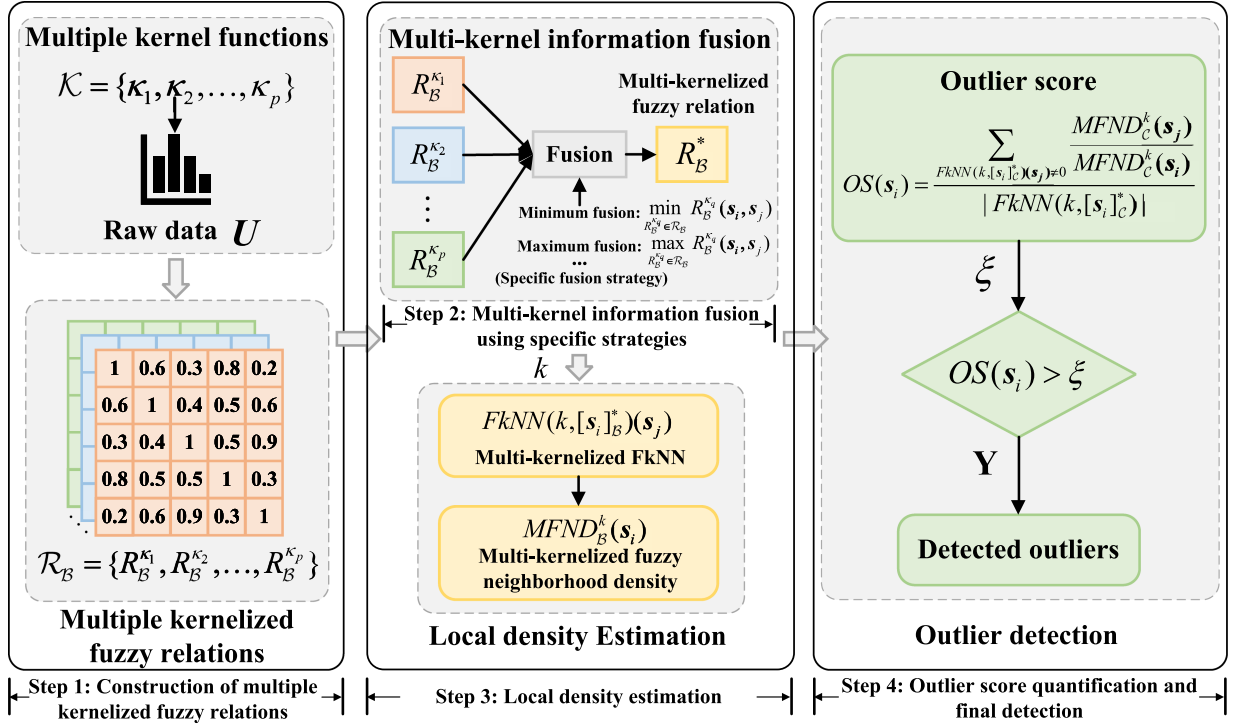


Fig. 1. The framework of MKFGOD.

detection method. Section IV proposes the Multi-Kernelized Fuzzy Granular Outlier Detector (MKFGOD) method. Section V validates the effectiveness of MKFGOD through extensive experiments. Section VI summarizes this study.

II. RELATED WORK

In this section, we review traditional outlier detection methods and fuzzy granular outlier detection methods.

A. Traditional Outlier Detection

Traditional outlier detection methods primarily encompass statistical, distance-based, and density-based methods, alongside more recent methods leveraging deep learning and ensemble learning.

Statistical methods typically assume specific data distributions. For instance, Yang et al. [25] employed a Gaussian mixture model to fit datasets, detecting outliers using a weighted sum of mixture proportions. Goldstein et al. [26] introduced a histogram-based method, achieving linear-time complexity for outlier scoring. Li et al. [27] utilized empirical cumulative distribution to detect outliers by estimating tail probabilities. However, these methods often perform poorly on data with unknown or complex distributions due to their reliance on distributional assumptions. Distance-based methods focus on inter-sample distances. Liu et al. [28] proposed the Isolation Forest-based method (IForest), which isolates samples using binary trees, identifying those farther from the root as outliers. Zhang et al. [29] developed a local distance-based method, detecting outliers based on a sample's distance relative to its neighbors. Angiulli et al. [30] introduced an efficient mechanism to enhance

the performance of SolvingSet-based methods. Nevertheless, these methods require pairwise distance computations, resulting in low efficiency and challenges with high-dimensional data. Density-based methods, represented by Breunig et al.'s [31] Local Outlier Factor (LOF), detect outliers by comparing a sample's relative density to its neighborhood. Tang et al. [32] introduced a local kernel density estimation method, using relative density to quantify outlier degrees. Li et al. [33] proposed a method based on the changing rate of directed density ratios, integrating distance, direction, and density information for outlier detection. Huang et al. [34] proposed a novel outlier detecting algorithm based on the outlier turning points, which uses natural neighbors to obtain the neighborhood parameter adaptively. However, these methods also depend on distance computations, inheriting the same limitations as distance-based methods in high-dimensional data.

In recent years, several methods based on deep learning, ensemble learning, and GBC have been proposed for outlier detection. For instance, Xu et al. [35] enhanced the traditional IForest by incorporating neural networks, introducing a deep isolation forest method. Song et al. [36] addressed fairness in unsupervised outlier detection, proposing a fair outlier detection method based on deep clustering. Cai et al. [10] improved traditional PCA with a learned robust PCA method, boosting outlier detection performance. Ouyang et al. [12] explored the integration of multiple outlier detection algorithms, where each selected detector focuses on identifying the most prominent outliers from its perspective. Zhang et al. [13] introduced a novel voting-based outlier detection method to mitigate the adverse effects of noisy data on classifier training. Sarvari et al. [37] proposed an ensemble method using boosting-based

autoencoders to enhance outlier detection performance. Furthermore, Su et al. [38] proposed an approach that fuses multi-granular-ball fuzzy information, effectively leveraging the descriptive power of diverse granularities to manage complex data uncertainty. In parallel, Su et al. [39] explored the strategy of mining granular-ball neighborhood information to enhance the detection of outliers. Nevertheless, these ensemble methods incur high computational costs due to the need for training, integrating, and tuning multiple models.

B. Fuzzy Granular Outlier Detection

Fuzzy granular outlier detection methods refer to methods built on FIG theory, which focus on leveraging uncertainty information associated with abnormal patterns to identify outliers in data. Generally, these methods are particularly effective for multivariate tabular data of small-to-medium sizes, where massive training data for deep learning is unavailable. Compared to established statistical methods, FIG is preferred when data boundaries are fuzzy or overlapping. For example, in a dataset with complex local irregularities, statistical methods might impose a global fit that misses local outliers. In contrast, FIG decomposes the data space into fine-grained fuzzy granules (local subsets). This allows the model to capture the local geometry and uncertainty of the data structure more accurately, identifying outliers that are indistinguishable using global rigid thresholds.

For example, Liu et al. [40] integrated FIG with Markov random walks, utilizing the stationary distribution derived from iterative computations to detect outliers. Yuan et al. [3] proposed an method based on FIG that fuses multi-fuzzy information granules to detect outliers. Chen et al. [41] proposed a consistency-guided method based on FIG for mixed-attribute data outlier detection. Gao et al. [42] developed a FIG-based method incorporating multi-scale granular-balls to detect various types of outliers. Yuan et al. [43] proposed a fuzzy neighborhood outlier detection method based on FIG, utilizing fuzzy neighborhood density to quantify the outlier degrees of samples. Su et al. [38] combined FIG with granular-ball computing, proposing a method that employs multi-granular-ball fuzzy information for outlier detection. Chen et al. [44] proposed a FIG-based method that integrates fuzzy approximations and relative entropy to address data uncertainty for effective outlier detection. Hu et al. [45] utilized FIG to exploit uncertainty information in three-way regions for detecting conditional outliers.

Although the above methods have achieved promising results, their reliance on simple linear techniques to model fuzzy and neighborhood relations during granulation overlooks complex nonlinear interactions between samples, leading to suboptimal detection performance. To address this, Wu et al. [20] incorporated Gaussian kernel functions into traditional fuzzy relations, defining kernelized fuzzy relations and constructing kernelized fuzzy rough sets to enhance outlier detection. However, the dependence on a single-kernel mapping limits the method's ability to adapt to varied and intricate data distributions, potentially causing the loss of key features during granulation and compromising outlier detection performance. Moreover, this method overlooks the benefits of multi-kernel information

fusion, which could enhance the modeling of fuzzy relations by integrating diverse kernel mappings to capture complex inter-sample relationships. Therefore, the integration of FIG with multi-kernel information fusion for outlier detection deserves further exploration.

III. PRELIMINARIES

In this section, we review the foundational concepts of FIG and kernel functions, and discuss the Detecting Fuzzy Neighborhood Outliers (DFNO) method, which is a recently proposed FIG-based method.

A. Fuzzy Information Granulation

Fuzzy Information Granulation (FIG) is a granulation method grounded in fuzzy binary relations, serving as a widely adopted strategy for information granulation. Let $U = [s_1; s_2; \dots; s_n] \in \mathbb{R}^{n \times m}$ denote the data matrix with n samples and m features (attributes), and $\mathcal{C} = \{c_1, c_2, \dots, c_m\}$ denote the conditional attribute set. For any $\mathcal{B} \subseteq \mathcal{C}$, $\mathcal{B}$ can induce a fuzzy relation $R_{\mathcal{B}}$ on U , which is defined as $R_{\mathcal{B}} : U \times U \rightarrow [0, 1]$. For any $s_i, s_j \in U$, if $R_{\mathcal{B}}$ satisfies 1) reflexivity: $R_{\mathcal{B}}(s_i, s_i) = 1$ and 2) symmetry: $R_{\mathcal{B}}(s_i, s_j) = R_{\mathcal{B}}(s_j, s_i)$, then $R_{\mathcal{B}}$ is called a fuzzy similarity relation. $R_{\mathcal{B}}$ can be denoted as a fuzzy relation matrix, i.e.,

$$M(R_{\mathcal{B}}) = \begin{pmatrix} r_{11}^{\mathcal{B}} & r_{12}^{\mathcal{B}} & \dots & r_{1n}^{\mathcal{B}} \\ r_{21}^{\mathcal{B}} & r_{22}^{\mathcal{B}} & \dots & r_{2n}^{\mathcal{B}} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1}^{\mathcal{B}} & r_{n2}^{\mathcal{B}} & \dots & r_{nn}^{\mathcal{B}} \end{pmatrix}, \quad (1)$$

where $r_{ij}^{\mathcal{B}} = R_{\mathcal{B}}(s_i, s_j) \in [0, 1]$ denotes the similarity between samples s_i and s_j .

For any $\mathcal{B} \subseteq \mathcal{C}$, the fuzzy relation $R_{\mathcal{B}}$ induced by $\mathcal{B}$ on U can granulate the data into a family set of fuzzy information granules, called a fuzzy granular structure. A fuzzy granular structure is formally denoted as

$$G(R_{\mathcal{B}}) = \{[s_1]_{\mathcal{B}}, [s_2]_{\mathcal{B}}, \dots, [s_n]_{\mathcal{B}}\}, \quad (2)$$

where $[s_i]_{\mathcal{B}} = \frac{R_{\mathcal{B}}(s_i, s_1)}{r_{s_1}^{\mathcal{B}}} + \frac{R_{\mathcal{B}}(s_i, s_2)}{r_{s_2}^{\mathcal{B}}} + \dots + \frac{R_{\mathcal{B}}(s_i, s_n)}{r_{s_n}^{\mathcal{B}}}$ denotes the fuzzy information granule determined by s_i with respect to $R_{\mathcal{B}}$, and $R_{\mathcal{B}}(s_i, s_j)$ denotes the similarity between s_i and s_j under $R_{\mathcal{B}}$. The cardinality of the fuzzy information granule $[s_i]_{\mathcal{B}}$ can be calculated by

$$|[s_i]_{\mathcal{B}}| = \sum_{j=1}^n r_{ij}^{\mathcal{B}}. \quad (3)$$

A fuzzy information granule $[s_i]_{\mathcal{B}}$ can be understood as a soft neighborhood centered at sample s_i . Unlike a traditional hard cluster or neighborhood that includes samples strictly within a fixed radius, a fuzzy information granule assigns a continuous membership degree (ranging from 0 to 1) to other samples based on their similarity to the center s_i . This structure is crucial because it models the local geometry and uncertainty of the data, allowing us to quantify to what extent samples are related, rather than making a rigid binary decision.

B. Kernel Functions

Kernel functions are a widely used feature transformation technique, commonly employed to address nonlinear separability issues in classifiers. They have been integrated with FIG to develop kernelized fuzzy rough sets [20], [46], [47]. According to Mercer's theorem, if a function κ is symmetric and positive-semidefinite, it implicitly defines a non-linear mapping $\phi : \mathcal{U} \rightarrow \mathcal{H}$ from the original input space to a high-dimensional Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}$. In this space, the kernel function corresponds to the inner product of the mapped feature vectors, i.e., $\kappa(s_i, s_j) = \langle \phi(s_i), \phi(s_j) \rangle_{\mathcal{H}}$. This implicit mapping is the core rationale for using kernels: it allows linear algorithms (such as measuring similarity or distance) to be performed in $\mathcal{H}$, thereby effectively capturing complex non-linear structures in the original space without the computational cost of explicit transformation.

According to [48], [49], any kernel $\kappa : \mathcal{U} \times \mathcal{U} \rightarrow [0, 1]$ with $\kappa(s_i, s_i) = 1$ is at least $T_{\cos}$ -transitive, where

$$T_{\cos}(p, q) = \max \left(pq - \sqrt{1-p^2} \sqrt{1-q^2}, 0 \right). \quad (4)$$

The concept of $T_{\cos}$ -transitivity is fundamental to our method. It guarantees that the fuzzy relations induced by valid kernels are not merely heuristic similarity measures but are formally fuzzy T -equivalence relations. This theoretical property ensures that the resulting fuzzy information granules are geometrically consistent and preserve the metric structure of the data, providing a rigorous mathematical foundation for the subsequent density estimation and outlier detection.

Since certain kernel functions satisfy reflexivity $\kappa(s_i, s_i) = 1$, symmetry $\kappa(s_i, s_j) = \kappa(s_j, s_i)$, and $T_{\cos}$ -transitive, the kernelized fuzzy binary relations derived from these functions constitute fuzzy $T_{\cos}$ -equivalence relations. For any samples $s_i, s_j \in \mathcal{U}$, we list several commonly used kernel functions that satisfy the above properties below [46].

- 1) Gaussian kernel: $\kappa_G(s_i, s_j) = \exp(-\frac{1}{\delta} \|s_i - s_j\|_2^2)$;
- 2) Exponential kernel: $\kappa_E(s_i, s_j) = \exp(-\frac{1}{\delta} \|s_i - s_j\|_2)$;
- 3) Rational quadratic kernel: $\kappa_R(s_i, s_j) = 1 - \frac{\|s_i - s_j\|_2^2}{\|s_i - s_j\|_2^2 + \delta}$.

The selection of these specific kernels is motivated by their complementary capabilities in capturing diverse data characteristics: the Gaussian kernel focuses on integrating local views of the data, whereas the Exponential kernel is well-suited for capturing global structures. The Rational quadratic kernel provides a multi-scale perspective, effectively acting as a scale-mixture of Gaussian kernels. By leveraging these kernel functions, we can replace the conventional fuzzy relations in FIG with kernelized fuzzy relations, thereby achieving kernelized fuzzy information granulation. The original fuzzy information granule and fuzzy granular structure will also be updated to kernelized fuzzy information granule and kernelized fuzzy granular structure. It is important to emphasize that integrating kernels into FIG is not merely a change of terminology. Traditional fuzzy relations often rely on distance metrics in the original input space, limiting their ability to model complex, non-linear distributions. In contrast, kernel functions implicitly map data into a high-dimensional feature space. By constructing fuzzy relations in

this space, we generate kernelized fuzzy information granules that can capture non-linear structural information inherent in the data—patterns that are otherwise indistinguishable using standard fuzzy relations in the original space.

C. Detecting Fuzzy Neighborhood Outliers

In this section, we review DFNO [43], a recently proposed outlier detection method based on FIG theory. DFNO integrates FIG theory with the classic density-based LOF method, initially granulating the dataset $\mathcal{U}$ into multiple fuzzy information granules using hybrid fuzzy similarity relations. Subsequently, it incorporates the k-Nearest Neighbors (kNN) concept into FIG theory, introducing Fuzzy kNN (FkNN), and calculates the density of samples through FKNN.

Definition 1: Given a positive integer k , for any $\mathcal{B} \subseteq \mathcal{C}$ and any $s_i, s_j \in \mathcal{U}$, the fuzzy neighborhood density of s_i is defined as

$$FND_{\mathcal{B}}^k(s_i) = \frac{1}{|[s_i]_{\mathcal{B}}^k|} \sum_{[s_i]_{\mathcal{B}}^k(s_j) \neq 0} RS_{\mathcal{B}}^k(s_i, s_j) \quad (5)$$

where $[s_i]_{\mathcal{B}}^k$ denotes the FkNN of s_i under the attribute subset $\mathcal{B}$ and $RS_{\mathcal{B}}^k(s_i, s_j)$ represents the fuzzy reachability similarity between samples.

Next, DFNO calculates the fuzzy neighborhood deviation degree as the outlier score of the sample by comparing the fuzzy neighborhood density between the sample and its neighborhood. Considering that calculating all attribute subsets in $\mathcal{C}$ is time-consuming and unrealistic, DFNO directly uses the full set of attributes $\mathcal{C}$ to calculate the outlier scores.

Definition 2: Given a positive integer k and any $s_i, s_j \in \mathcal{U}$, the outlier score of s_i is defined as

$$FNDD_{\mathcal{C}}^k(s_i) = \frac{1}{|[s_i]_{\mathcal{C}}^k|} \sum_{[s_i]_{\mathcal{C}}^k(s_j) \neq 0} \frac{FND_{\mathcal{C}}^k(s_j)}{FND_{\mathcal{C}}^k(s_i)}. \quad (6)$$

Although DFNO achieves promising detection results, it defines fuzzy relations between samples in a simplistic linear manner, overlooking complex nonlinear interactions. To address this, we first integrate kernel functions into FIG theory and propose a novel multi-kernel information fusion model to construct multi-kernelized fuzzy relations, enhancing the modeling of traditional fuzzy relations and improving outlier detection performance.

IV. MULTI-KERNELIZED FUZZY GRANULAR OUTLIER DETECTOR

In this section, we first propose a novel multi-kernel information fusion model to construct multi-kernelized fuzzy relations. Subsequently, based on the multi-kernelized fuzzy information granules obtained through granulation, we introduce the Multi-Kernelized Fuzzy Granular Outlier Detector (MKFGOD).

A. Multi-Kernel Information Fusion Model

As mentioned above, although DFNO achieves promising detection results, it defines fuzzy relations between samples in a simplistic linear manner, thereby overlooking complex

nonlinear interactions and limiting its performance in diverse datasets. To address this issue, we enhance the capability of traditional fuzzy relations in capturing complex inter-sample relationships by constructing a novel multi-kernel information fusion model with diverse fusion strategies.

Specifically, given a set of kernel functions $\mathcal{K} = \{\kappa_1, \kappa_2, \dots, \kappa_p\}$, for any kernel function $\kappa_q \in \mathcal{K}$ and $\mathcal{B} \subseteq \mathcal{C}$, one can define a kernelized fuzzy relation $R_B^{\kappa_q}$ induced by $\mathcal{B}$ to characterize the relationships between samples. Consequently, this yields a set of multiple kernelized fuzzy relations

$$\mathcal{R}_B = \{R_B^{\kappa_1}, R_B^{\kappa_2}, \dots, R_B^{\kappa_p}\}. \quad (7)$$

For any $R_B^{\kappa_q} \in \mathcal{R}_B$, the dataset $\mathcal{U}$ can be granulated into a kernelized fuzzy granular structure

$$G(R_B^{\kappa_q}) = \{[s_1]_B^{\kappa_q}, [s_2]_B^{\kappa_q}, \dots, [s_n]_B^{\kappa_q}\}, \quad (8)$$

where $[s_i]_B^{\kappa_q} = \frac{R_B^{\kappa_q}(s_i, s_1)}{s_1} + \frac{R_B^{\kappa_q}(s_i, s_2)}{s_2} + \dots + \frac{R_B^{\kappa_q}(s_i, s_n)}{s_n}$ denotes the kernelized fuzzy information granule.

Effectively integrating kernelized fuzzy relations defined by multiple kernel functions is crucial for multi-kernel information fusion. In this study, we propose five fusion methods to calculate the multi-kernelized fuzzy relations, including minimum fusion, maximum fusion, mean fusion, multiply fusion, and weighted fusion. These strategies are selected based on fundamental aggregation principles in fuzzy set theory and information fusion, providing diverse logical perspectives to manage data uncertainty and capture complex inter-sample relationships. Specifically, minimum fusion implements a conservative consensus logic that reduces false positives by requiring similarity across all kernels, whereas maximum fusion employs an inclusive logic to preserve the strongest relational link identified by any individual kernel mapping. To mitigate individual kernel bias, mean fusion provides a balanced and smoothed estimation. Furthermore, multiply fusion follows the T-norm principle in fuzzy logic, acting as a rigorous soft intersection to sharpen the distinction between similar and dissimilar pairs. Finally, weighted fusion serves as an adaptive mechanism that assigns higher importance to kernels with stronger discriminative capability. The mathematical definitions are given as follows.

- 1) *Minimum fusion*: $R_B^*(s_i, s_j) = \min_{R_B^{\kappa_q} \in \mathcal{R}_B} R_B^{\kappa_q}(s_i, s_j)$;
- 2) *Maximum fusion*: $R_B^*(s_i, s_j) = \max_{R_B^{\kappa_q} \in \mathcal{R}_B} R_B^{\kappa_q}(s_i, s_j)$;
- 3) *Mean fusion*: $R_B^*(s_i, s_j) = \frac{1}{|\mathcal{R}_B|} \sum_{R_B^{\kappa_q} \in \mathcal{R}_B} R_B^{\kappa_q}(s_i, s_j)$;
- 4) *Multiply fusion*: $R_B^*(s_i, s_j) = \prod_{R_B^{\kappa_q} \in \mathcal{R}_B} R_B^{\kappa_q}(s_i, s_j)$.

Additionally, considering that different kernel functions may capture distinct aspects of the data and thus contribute unequally to the representation, it is essential to assign different weights to them. Fuzzy information entropy, as an important uncertainty measure in FIG theory, can quantitatively reflect the discriminative capability of a fuzzy relation [3], [38]. We introduce kernelized fuzzy relations into fuzzy information entropy and construct the following kernelized fuzzy information entropy.

Definition 3: For any $R_B^{\kappa_q} \in \mathcal{R}_B$, $G(R_B^{\kappa_q}) = \{[s_1]_B^{\kappa_q}, [s_2]_B^{\kappa_q}, \dots, [s_n]_B^{\kappa_q}\}$ is the set of kernelized fuzzy information granules granulated by $R_B^{\kappa_q}$, the kernelized fuzzy

information entropy w.r.t. $R_B^{\kappa_q}$ is defined as

$$KEN(R_B^{\kappa_q}) = -\frac{1}{n} \sum_{s_i \in \mathcal{U}} \log_2 \frac{|[s_i]_B^{\kappa_q}|}{n}, \quad (9)$$

where $|[s_i]_B^{\kappa_q}| = \sum_{s_j \in \mathcal{U}} R_B^{\kappa_q}(s_i, s_j)$ denotes the cardinality of $[s_i]_B^{\kappa_q}$.

The kernelized fuzzy information entropy provides an objective, data-driven mechanism to quantify the discriminative capability of fuzzy relations. A larger entropy indicates a finer granulation where the fuzzy information granules are smaller and more compact, thereby better preserving the local geometric details of the data manifold to identify outliers. Conversely, lower entropy implies coarse granules that may obscure structural details. By assigning higher weights to kernels with larger entropy, MKFGOD adaptively prioritizes the most informative relations without manual tuning, ensuring that the fused relation captures the maximum amount of discriminative information across diverse datasets. Finally, the weighted fusion is defined as

$$R_B^*(s_i, s_j) = \sum_{R_B^{\kappa_q} \in \mathcal{R}_B} w_B^{\kappa_q} R_B^{\kappa_q}(s_i, s_j), \quad (10)$$

where $w_B^{\kappa_q} = \frac{KEN(R_B^{\kappa_q})}{\sum_{R_B^{\kappa_r} \in \mathcal{R}_B} KEN(R_B^{\kappa_r})}$ denotes the weight coefficient.

It is essential to ensure that the fusion process preserves the fundamental properties of fuzzy similarity relations. Let $R_B^{\kappa_q}$ be the kernelized fuzzy relation derived from any kernel $\kappa_q \in \mathcal{K}$. Since each $R_B^{\kappa_q}$ satisfies reflexivity ($R(s_i, s_i) = 1$) and symmetry ($R(s_i, s_j) = R(s_j, s_i)$), their aggregation via the proposed five fusion strategies (min, max, mean, multiply, and weighted sum) inherently maintains these properties. Specifically, for any $s_i, s_j \in \mathcal{U}$, $R_B^*(s_i, s_i) = \text{aggregate}(1, \dots, 1) = 1$, and $R_B^*(s_i, s_j) = \text{aggregate}(R_B^{\kappa_1}(s_i, s_j), \dots) = R_B^*(s_j, s_i)$. Furthermore, the fusion process acts as a structural stabilizer; while individual kernels capture specific non-linear structures, their integration mitigates the inductive bias of a fixed single-kernel mapping, leading to more consistent fuzzy information granules.

Thus far, we propose a multi-kernel information fusion model incorporating various fusion strategies. Next, based on this model, we will introduce the novel method, MKFGOD.

B. Detection Model

Based on the multi-kernelized fuzzy relation R_B^* introduced above, the dataset $\mathcal{U}$ can be granulated into a multi-kernelized fuzzy granular structure

$$G(R_B^*) = \{[s_1]_B^*, [s_2]_B^*, \dots, [s_n]_B^*\}, \quad (11)$$

where $[s_i]_B^* = \frac{R_B^*(s_i, s_1)}{s_1} + \frac{R_B^*(s_i, s_2)}{s_2} + \dots + \frac{R_B^*(s_i, s_n)}{s_n}$ denotes the multi-kernelized fuzzy information granule.

Definition 4: Given a positive integer k , for any $\mathcal{B} \subseteq \mathcal{C}$ and $s_i, s_j \in \mathcal{U}$, the multi-kernelized FkNN of s_i is defined as

$$[s_i]_{\mathcal{B}}^{*k}(s_j) = \begin{cases} R_{\mathcal{B}}^*(s_i, s_j), & \text{if } R_{\mathcal{B}}^*(s_i, s_j) \geq MKS_{\mathcal{B}}^k(s_i); \\ 0, & \text{elsewhere,} \end{cases} \quad (12)$$

where $MKS_{\mathcal{B}}^k(s_i)$ denotes the multi-kernelized k -similarity of s_i , which needs to satisfy the following two conditions

- 1) At least k samples $s_j \in \mathcal{U} - \{s_i\}$ satisfy $R_{\mathcal{B}}^*(s_i, s_j) \geq MKS_{\mathcal{B}}^k(s_i)$;
- 2) At most $k-1$ samples $s_j \in \mathcal{U} - \{s_i\}$ satisfy $R_{\mathcal{B}}^*(s_i, s_j) > MKS_{\mathcal{B}}^k(s_i)$.

$MKS_{\mathcal{B}}^k(s_i)$ corresponds to the k -distance concept in the LOF algorithm [31]. While LOF uses distance to define the boundary of a neighborhood, we utilize similarity. Thus, the condition $R_{\mathcal{B}}^*(s_i, s_j) \geq MKS_{\mathcal{B}}^k(s_i)$ is the direct dual of the LOF neighborhood condition.

Furthermore, to reduce the statistical fluctuations caused by directly calculating multi-kernelized fuzzy similarity, we introduce the multi-kernelized fuzzy reachability similarity to enhance the subsequent density measurement, thereby improving outlier detection performance.

Definition 5: Given a positive integer k , for any $\mathcal{B} \subseteq \mathcal{C}$ and any $s_i, s_j \in \mathcal{U}$, the multi-kernelized fuzzy reachability similarity of s_i w.r.t. s_j is defined as

$$MRS_{\mathcal{B}}^k(s_i, s_j) = \min\{MKS_{\mathcal{B}}^k(s_j), R_{\mathcal{B}}^*(s_i, s_j)\}. \quad (13)$$

Similarly, $MRS_{\mathcal{B}}^k(s_i, s_j)$ corresponds to the reachability-distance in LOF. LOF uses the maximum of k -distance and actual distance to smooth density estimates. In our similarity space, this transforms into taking the minimum operation. This mechanism is critical for preventing numerical instability and reducing statistical fluctuations. Specifically, in the presence of duplicate or extremely close samples (where $R^* \approx 1$), utilizing raw similarity would artificially inflate the local density. By taking the minimum with MKS , MRS effectively caps the similarity contribution of these neighbors, ensuring that the density estimate reflects the overall structural tightness of the k -neighborhood rather than being skewed by singular proximal pairs.

We introduce multi-kernelized fuzzy relations to refine the original fuzzy neighborhood density, thereby improving the density measurement.

Definition 6: Given a positive integer k , for any $\mathcal{B} \subseteq \mathcal{C}$ and any $s_i, s_j \in \mathcal{U}$. Let $[s_i]_{\mathcal{B}}^{*k}$ be the multi-kernelized FkNN of sample s_i . We first define the fuzzy cardinality of this set as

$$|[s_i]_{\mathcal{B}}^{*k}| = \sum_{[s_i]_{\mathcal{B}}^{*k}(s_j) \neq 0} R_{\mathcal{B}}^*(s_i, s_j). \quad (14)$$

Based on this, the multi-kernelized fuzzy neighborhood density of s_i is defined as

$$MFND_{\mathcal{B}}^k(s_i) = \frac{1}{|[s_i]_{\mathcal{B}}^{*k}|} \sum_{[s_i]_{\mathcal{B}}^{*k}(s_j) \neq 0} MRS_{\mathcal{B}}^k(s_i, s_j). \quad (15)$$

Finally, we calculate the outlier score of a sample by comparing its multi-kernelized fuzzy neighborhood density with

that of its neighborhood to reflect the outlier degree of the sample. All the definitions introduced above are considered under different attribute subsets $\mathcal{B} \subseteq \mathcal{C}$. However, in practical applications, exhaustively calculating all attribute subsets is infeasible. Therefore, we directly use the full attribute set $\mathcal{C}$ to calculate the outlier score.

Definition 7: Given a positive integer k and any $s_i, s_j \in \mathcal{U}$, the outlier score of s_i is defined as

$$OS(s_i) = \frac{1}{|[s_i]_{\mathcal{C}}^{*k}|} \sum_{[s_i]_{\mathcal{C}}^{*k}(s_j) \neq 0} \frac{MFND_{\mathcal{C}}^k(s_j)}{MFND_{\mathcal{C}}^k(s_i)}. \quad (16)$$

In the above definition, if the multi-kernelized fuzzy neighborhood density of sample s_i is significantly lower than that of its neighborhood s_j , it indicates that the local density around s_i is smaller. Consequently, the calculated outlier score is higher, suggesting a greater likelihood that s_i is an outlier. After calculating the outlier score for each sample, we can detect specific outliers by setting an outlier detection threshold ξ . If a sample's outlier score exceeds ξ , it is identified as an outlier by the MKFGOD.

C. Detection Algorithm

Based on the proposed MKFGOD method, we present the corresponding outlier detection algorithm below. As shown in Algorithm 1, the algorithm requires the following inputs, including data matrix $\mathcal{U}$, conditional attribute set $\mathcal{C}$, kernel function set $\mathcal{K}$, specific fusion strategy, and positive integer k . MKFGOD first constructs the multi-kernelized fuzzy relations $R_{\mathcal{C}}^*(s_i, s_j)$ based on the input multiple kernel functions $\mathcal{K}$ and the specified fusion strategy, including minimum fusion, maximum fusion, mean fusion, multiply fusion, and weighted fusion. Next, MKFGOD calculates each sample's multi-kernelized k -similarity $MKS_{\mathcal{C}}^k(s_i)$ and multi-kernelized FkNN $[s_i]_{\mathcal{C}}^{*k}$. To reduce statistical fluctuations during density estimation, MKFGOD calculates the multi-kernelized fuzzy reachability similarity $MRS_{\mathcal{C}}^k(s_i, s_j)$ between samples before performing density estimation. Then, it estimates each sample's multi-kernelized fuzzy neighborhood density $MFND_{\mathcal{C}}^k(s_i)$. Finally, the outlier score $OS(s_i)$ is calculated by comparing the density of a sample with that of its multi-kernelized FkNN.

In terms of time complexity, calculating the multi-kernelized fuzzy relations, k -similarity, and FkNN requires a time complexity of $O(n^2)$. Subsequently, calculating the multi-kernelized fuzzy reachability similarity and fuzzy neighborhood density also incurs a time complexity of $O(n^2)$. Finally, calculating the outlier score has a time complexity of $O(n)$. Therefore, the overall time complexity of the MKFGOD algorithm is $O(n^2)$. While this is manageable for small-to-medium datasets, scalability for ultra-large datasets can be further enhanced by incorporating approximation techniques like the Nyström method [50] or Random Fourier Features [51] to achieve near-linear complexity. Additionally, leveraging GBC [52] to represent massive data with a reduced number of granular-balls to reduce the effective input size can significantly improve the practical utility of the proposed method.

Algorithm 1: MKFGOD Algorithm.

Input: Data matrix U , conditional attribute set $\mathcal{C}$, kernel function set $\mathcal{K}$, fusion strategy, positive integer k

Output: Outlier score OS

```

1 for  $s_i \in U$  do
2   for  $s_j \in U$  do
3     Calculate the multi-kernelized fuzzy relations
        $R_{\mathcal{C}}^*(s_i, s_j)$  with kernel functions  $\mathcal{K}$  and input
       fusion strategy;
4   end
5   Calculate the multi-kernelized k-similarity
        $MKS_{\mathcal{C}}^k(s_i)$  of  $s_i$  by Eq. (12);
6   Calculate the multi-kernelized FkNN  $[s_i]_{\mathcal{B}}^{*k}$  of  $s_i$ 
       by Eq. (12);
7 end
8 for  $s_i \in U$  do
9   for  $s_j \in U$  do
10    Calculate the multi-kernelized fuzzy
        reachability similarity  $MRS_{\mathcal{C}}^k(s_i, s_j)$  of  $s_i$ 
        w.r.t.  $s_j$  by Eq. (13);
11   end
12   Calculate the multi-kernelized fuzzy neighborhood
        density  $MFND_{\mathcal{C}}^k(s_i)$  of  $s_i$  by Eq. (15);
13 end
14 for  $s_i \in U$  do
15   Calculate the outlier score  $OS(s_i)$  of  $s_i$  by
       Eq. (16);
16 end
17 return  $OS$ .

```

V. EXPERIMENTAL DESIGN AND RESULT ANALYSIS

In this section, we validate the effectiveness of MKFGOD through extensive experiments. We first introduce the experimental datasets and evaluation metrics used. Then, we compare the experimental results and conduct an in-depth analysis.

A. Experimental Dataset

We select 24 datasets commonly used for outlier detection experiments, sourced from relevant public web pages.¹ The basic information of these datasets is shown in Table I, with sample sizes ranging from 101 to 7,200, attribute counts from 6 to 58, outlier counts from 4 to 573, and outlier ratios from 0.4% to 29.7%. Prior to the experiments, the datasets are normalized using the min-max method, and label information is removed to create an unsupervised outlier detection environment.

B. Comparative Experimental Setup

In the experiments, we select 17 outlier detection methods, including both recently proposed and classic methods, such as the local Granular-Ball Density-based Outlier (GBDO) detection [17], Granular-Ball computing-based method using

Random walk for Anomaly Detection (GBRAD) [53], Multi-scale Granular Balls-based Outlier Detection (MGBOD) [42], Anomaly Detection based on improved k-nearest Neighbor Rough Sets (ADkNRS) [54], Granular-Ball Mean-shift Outlier Detector (GBMOD) [55], Granular-Ball Fuzzy Rough sets-based anomaly Detection (GBFRD) [4], Weighted Fuzzy-Rough Density-based Anomaly detection (WFRDA) [19], Multi-Fuzzy Granules Anomaly Detection (MFGAD) [3], Incomplete Local and Global Neighborhood Information (ILGNI) [56], Deep Isolation Forest for anomaly detection (DIF) [35], Rotation-based Outlier Detection (ROD) [57], Empirical Cumulative distribution functions-based Outlier Detection (ECOD) [27], Directed density ratio Changing Rate-based Outlier Detection (DCROD) [33], COPula-based Outlier Detection (COPOD) [58], Deep Support Vector Data Description-based outlier detection (DeepSVDD) [11], Isolation Forest-based outlier detection (IForest) [28], and Local Outlier Factor-based outlier detection (LOF) [31].

To ensure the fairness and reproducibility of the experiments, the competing methods were implemented using standard open-source libraries and official codebases. Specifically, classic methods (e.g., LOF, IForest, COPOD, DeepSVDD) were executed using the PyOD (Python Outlier Detection) library [59], which provides robust and unified implementations. For recently proposed state-of-the-art methods (e.g., GBDO, GBRAD, MGBOD), we utilized the official source codes provided by the respective authors via their public GitHub repositories.

Among the aforementioned methods, most involve hyperparameters that require tuning based on the dataset. To ensure experimental fairness, we set reasonable tuning ranges for these methods based on existing studies. For GBDO and GBMOD, the number of granular-ball neighborhoods is tuned within the range $[2, \min(60, |\mathcal{GB}|)]$ with a step size of 1, where $|\mathcal{GB}|$ denotes the number of granular-balls generated on the dataset. For GBRAD, the granular-ball splitting parameter is tuned within the range $[3, 20]$ with a step size of 1. For MGBOD, WFRDA, and MFGAD, the fuzzy radius size is tuned within the range $[0.1, 2.0]$ with a step size of 0.1. For ADkNRS, DCROD, and LOF, the number of neighborhood samples is tuned within the range $[2, 60]$ with a step size of 1. For GBFRD, the fuzzy radius size is tuned within the range $[0, 1]$ with a step size of 0.05. The number of base estimators in IForest is set to 100. In our proposed MKFGOD method, the set of kernel functions is $\mathcal{K} = \{\text{Gaussian, Exponential, Rational quadratic}\}$. To ensure a rigorous and fair evaluation, the optimal fusion strategy and the neighborhood parameter k are determined through a systematic grid search procedure. Specifically, the fusion strategy is selected from the five proposed candidates (minimum, maximum, mean, multiply, and weighted fusion), while k is tuned within the range $[2, 60]$ with a step size of 1. This approach allows the model to adaptively identify the most suitable configuration for diverse data distributions, which is a standard practice in hyperparameter optimization. To ensure the robustness of MKFGOD across diverse data scales, the kernel parameter δ is determined in a self-adaptive manner rather than through manual tuning. Specifically, δ is defined as the median of all non-diagonal elements in the distance matrix, which accounts for both numerical and nominal

¹<https://github.com/BELLoney/Outlier-detection>

TABLE I
DETAILS OF THE 24 DATASETS USED IN THE EXPERIMENTS

No.	Datasets	Abbr.	# Samples	# Attributes	# Outliers	% Outlier ratios	Domains
1	Breast_cancer_variant1	Breast	286	9	85	29.7%	Health and Medicine
2	Chess_nowin_34_variant1	Chess1	1703	36	34	2.0%	Games
3	Chess_nowin_87_variant1	Chess2	1756	36	87	5.0%	Games
4	Lymphography	Lym	148	8	6	4.1%	Health and Medicine
5	Monks_0_12_variant1	Monks1	240	6	12	5.0%	Social Science
6	Monks_0_25_variant1	Monks2	253	6	25	9.9%	Social Science
7	Monks_0_4_variant1	Monks3	232	6	4	1.7%	Social Science
8	Mushroom_p_365_variant1	Mush1	4573	22	365	8.0%	Biology
9	Mushroom_p_467_variant1	Mush2	4675	22	467	10.0%	Biology
10	Mushroom_p_573_variant1	Mush3	4781	22	573	12.0%	Biology
11	Tic_tac_toe_negative_32_variant1	Tic1	658	9	32	4.9%	Games
12	Tic_tac_toe_negative_54_variant1	Tic2	680	9	54	7.9%	Games
13	Tic_tac_toe_negative_69_variant1	Tic3	695	9	69	9.9%	Games
14	Zoo_variant1	Zoo	101	16	17	16.8%	Biology
15	Annthyroid	Ann	7200	6	534	7.4%	Health and Medicine
16	Letter	Letter	1600	32	100	6.3%	Computer Science
17	Pageblocks_1_258_variant1	Page	5171	11	258	5.0%	Computer Science
18	Satimage2	Sat	5803	36	71	1.2%	Earth Science
19	Spambase_spam_56_variant1	Spam	2844	58	56	2.0%	Computer Science
20	Vowels	Vow	1456	12	50	3.4%	Computer Science
21	Waveform_0_100_variant1	Wave	3443	21	100	2.9%	Physics and Chemistry
22	Yeast_ERL_5_variant1	Yeast	1141	8	5	0.4%	Biology
23	Abalone_variant1	Aba	4177	9	79	1.9%	Biology
24	Sick_sick_35_variant1	Sick	3576	29	35	1.0%	Health and Medicine

TABLE II
THE MAIN IDEAS AND HYPERPARAMETER SETTINGS OF ALL METHODS IN THE EXPERIMENT, WHERE $\times$ DENOTES THAT THE METHOD DOES NOT REQUIRE HYPERPARAMETER SELECTION OR STEP SIZE

Methods (Years)	Main ideas	Hyperparameter tuning ranges	Step sizes
MKFGOD (Ours)	Multi-kernelized fuzzy granular outlier detection	five fusion strategies, $k \in [2, 60]$	one by one, 1
GBDO (2025) [17]	Local granular-ball density	$[2, \min(60, GB)]$	1
GBRAD (2025) [53]	Granular-ball random walk	$[3, 20]$	1
MGBOD (2025) [42]	Multi-scale granular-balls	$[0.1, 2.0]$	0.1
ADkNRS (2025) [54]	Improved k -nearest neighbor rough sets	$[2, 60]$	1
GBMOD (2025) [55]	Granular-ball mean-shift	$[2, \min(60, GB)]$	1
GBFRD (2024) [4]	Granular-ball fuzzy approximation accuracy	$[0, 1]$	0.05
WFRDA (2023) [19]	Weighted fuzzy-rough density	$[0.1, 2.0]$	0.1
MFGAD (2023) [3]	Multi-fuzzy granules	$[0.1, 2.0]$	0.1
ILGNI (2023) [56]	Incomplete local and global neighborhood information	$\times$	$\times$
DIF (2023) [35]	Deep isolation forest	$\times$	$\times$
ROD (2022) [57]	Rotation information	$\times$	$\times$
ECOD (2022) [27]	Empirical cumulative distribution functions	$\times$	$\times$
DCROD (2022) [33]	Directed density ratio changing rate	$[2, 60]$	1
COPOD (2020) [58]	Copula functions	$\times$	$\times$
DeepSVDD (2018) [11]	Deep support vector data description	$\times$	$\times$
IForest (2008) [28]	Isolation forest	100	$\times$
LOF (2000) [31]	Local density	$[2, 60]$	1

attributes by combining Euclidean and Hamming distances. This median-based heuristic allows the kernels to adaptively characterize the local density and fuzzy similarity relations according to the inherent distribution of the dataset, effectively enhancing the model's flexibility in real-world applications. The main ideas and hyperparameter settings for all methods are summarized in Table II.

To comprehensively evaluate the detection performance, we employ the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) as the primary metrics.

C. Experimental Results and Analysis

1) *Effectiveness Analysis*: As illustrated in Fig. 2, the MKFGOD's ROC curves are closer to the top-left corner of the coordinate axes on the Breast, Chess1, Chess2, Lym, Mush1,

Mush2, Mush3, Tic1, Tic2, Tic3, Zoo, Letter, and Vow datasets, indicating superior performance of our method.

To further validate the effectiveness of MKFGOD quantitatively, we present the AUC results in Table III. MKFGOD achieves the highest AUC score in 17 out of 24 datasets and obtains the best average AUC of 0.942, surpassing the second-best method by 8.6%. This demonstrates that MKFGOD consistently achieves relatively strong detection performance across datasets of varying scales.

The ROC curves and AUC results presented above confirm the effectiveness of MKFGOD. Next, we will conduct a deeper analysis and discussion of MKFGOD through statistical analysis and hyperparameter sensitivity analysis.

2) *Hyperparameter Sensitivity Analysis*: In the proposed MKFGOD method, the hyperparameter k , representing the number of neighborhood samples, requires tuning based on the

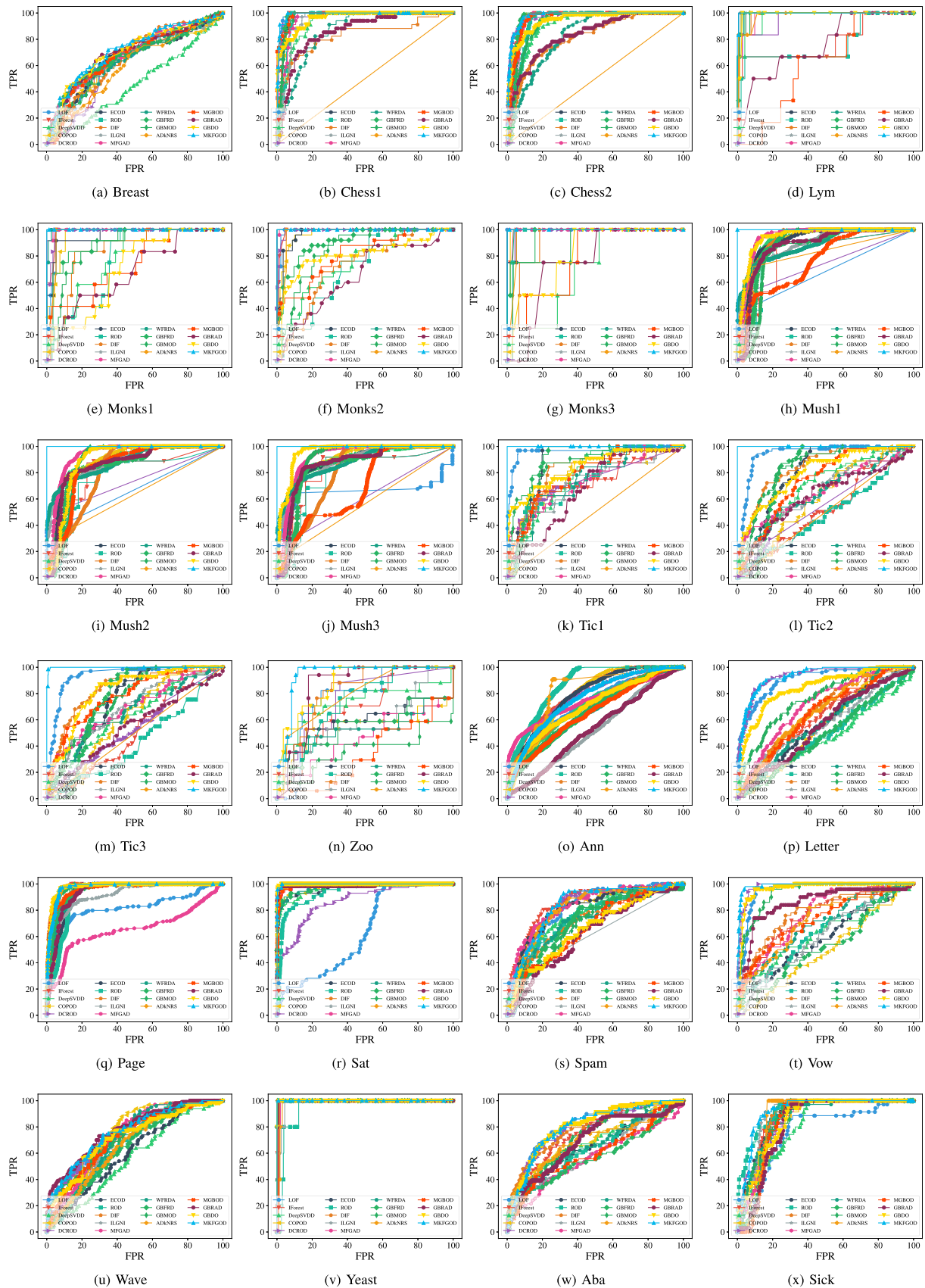


Fig. 2. ROC curves of 18 methods and 24 datasets.

TABLE III
AUC RESULTS COMPARISON OF 18 OUTLIER DETECTION METHODS ACROSS 24 DATASETS. THE BEST AUC RESULT IN EACH ROW IS **BOLDED**.

Datasets	LOF	IForest	DeepSVDD	COPOD	DCROD	ECOD	ROD	DIF	ILGNI	MFGAD	WFRDA	GBFRD	GBMOD	ADkNRS	MGBOD	GBRAD	GBDO	MKFGOD
Breast	0.654	0.635	0.458	0.632	0.604	0.656	0.692	0.654	0.659	0.685	0.654	0.671	0.644	0.594	0.664	0.695	0.701	0.717
Chess1	0.987	0.973	0.929	0.967	0.979	0.968	0.954	0.830	0.935	0.957	0.862	0.969	0.968	0.500	0.987	0.860	0.961	0.990
Chess2	0.968	0.957	0.908	0.951	0.970	0.952	0.930	0.814	0.941	0.952	0.828	0.953	0.921	0.500	0.955	0.834	0.944	0.977
Lym	0.994	0.781	0.969	0.994	0.954	0.996	0.762	0.982	0.991	0.978	0.993	0.987	0.986	0.979	0.612	0.757	0.989	0.999
Monks1	0.973	0.984	0.781	0.959	0.986	0.957	0.762	0.837	1.000	0.997	0.996	0.906	0.870	0.967	0.726	0.671	0.696	1.000
Monks2	0.989	0.988	0.752	0.967	0.992	0.969	0.722	0.724	1.000	0.999	1.000	0.910	0.852	0.969	0.814	0.653	0.806	1.000
Monks3	1.000	0.992	0.703	0.991	0.989	0.987	0.968	0.912	1.000	1.000	1.000	0.956	0.886	0.967	0.784	0.754	0.816	1.000
Mush1	0.682	0.812	0.909	0.902	0.728	0.900	0.821	0.929	0.894	0.942	0.886	0.860	0.858	0.808	0.757	0.872	0.943	1.000
Mush2	0.672	0.784	0.888	0.888	0.710	0.867	0.761	0.814	0.885	0.930	0.885	0.841	0.871	0.619	0.841	0.860	0.885	1.000
Mush3	0.657	0.805	0.871	0.884	0.604	0.866	0.779	0.800	0.882	0.915	0.882	0.840	0.897	0.559	0.689	0.860	0.953	1.000
Tic1	0.964	0.721	0.772	0.785	0.723	0.878	0.749	0.801	0.722	0.737	0.758	0.881	0.877	0.500	0.789	0.681	0.838	1.000
Tic2	0.943	0.523	0.695	0.615	0.548	0.790	0.483	0.839	0.646	0.670	0.649	0.803	0.845	0.500	0.739	0.621	0.800	1.000
Tic3	0.935	0.506	0.633	0.617	0.536	0.724	0.430	0.830	0.689	0.690	0.660	0.745	0.757	0.500	0.781	0.547	0.818	0.999
Zoo	0.769	0.690	0.727	0.550	0.752	0.582	0.735	0.811	0.589	0.466	0.526	0.539	0.363	0.791	0.342	0.853	0.880	0.941
Ann	0.718	0.802	0.710	0.776	0.757	0.789	0.860	0.707	0.522	0.746	0.658	0.658	0.665	0.809	0.652	0.531	0.700	0.761
Letter	0.906	0.630	0.457	0.569	0.928	0.572	0.493	0.674	0.582	0.711	0.584	0.488	0.764	0.656	0.648	0.558	0.841	0.934
Page	0.824	0.970	0.948	0.930	0.953	0.938	0.921	0.980	0.914	0.655	0.944	0.951	0.974	0.976	0.961	0.937	0.979	0.975
Sat	0.637	0.994	0.961	0.974	0.865	0.965	0.933	0.997	0.994	0.989	0.972	0.971	0.998	0.994	0.988	0.991	0.999	0.998
Spam	0.800	0.846	0.756	0.815	0.825	0.812	0.736	0.718	0.607	0.836	0.725	0.715	0.696	0.819	0.667	0.624	0.661	0.833
Vow	0.951	0.761	0.621	0.496	0.978	0.593	0.626	0.786	0.609	0.767	0.686	0.533	0.931	0.717	0.731	0.857	0.963	0.989
Wave	0.741	0.708	0.564	0.734	0.743	0.608	0.670	0.714	0.705	0.736	0.704	0.648	0.720	0.670	0.719	0.778	0.719	0.766
Yeast	0.992	0.997	0.997	0.997	0.990	0.995	0.949	0.987	0.981	0.992	0.998	1.000	0.999	0.984	0.991	0.997	1.000	0.999
Aba	0.811	0.753	0.677	0.696	0.780	0.653	0.629	0.787	0.604	0.565	0.661	0.576	0.717	0.689	0.603	0.689	0.800	0.799
Sick	0.767	0.881	0.815	0.839	0.875	0.883	0.927	0.879	0.902	0.860	0.868	0.855	0.877	0.865	0.837	0.824	0.860	0.925
Average	0.847	0.812	0.771	0.814	0.824	0.829	0.762	0.825	0.802	0.824	0.807	0.802	0.831	0.747	0.762	0.763	0.856	0.942

characteristics of different datasets. To investigate MKFGOD's sensitivity to k , we plot the AUC performance curves of MKFGOD across all datasets as a function of varying k values. Experiments reveal that different fusion strategies impact AUC results for certain k values but do not alter the overall AUC trend with varying k . Therefore, we focus on the maximum fusion strategy in the hyperparameter sensitivity analysis.

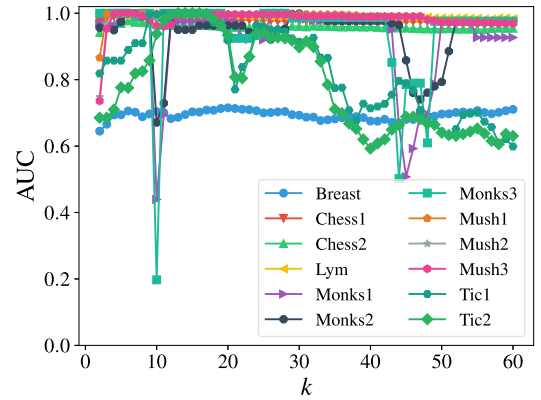
As shown in Fig. 3, the AUC scores of MKFGOD on most datasets gradually increase with larger k values before stabilizing. This suggests that expanding the neighborhood size enhances the model's ability to distinguish outliers from inliers by improving density estimation. However, when the neighborhood becomes too large, the density estimates may become overly generalized, dominated by the global structure of the dataset, which diminishes the model's sensitivity to local variations and results in stable performance.

On the Tic1, Tic2, and Tic3 datasets, MKFGOD's AUC scores initially rise with increasing k , then decline and eventually stabilize. The initial increase indicates that a moderately larger neighborhood enhances outlier detection performance by capturing more representative local density information. The subsequent decline and stabilization suggest that an excessively large neighborhood may dilute critical local density distinctions, reducing the model's effectiveness.

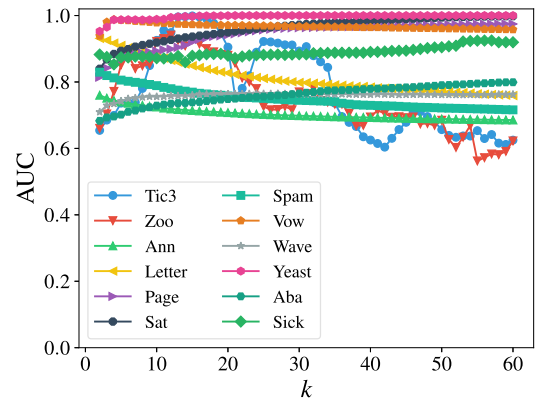
Overall, MKFGOD demonstrates low sensitivity to the choice of k across most datasets, particularly within the stable range of $[10, 30]$ as illustrated in Fig. 3. This stability stems from the multi-kernel fusion mechanism, which mitigates the impact of local density fluctuations. We recommend setting the default value of k to 15, as it provides a robust balance between capturing fine-grained local structures and suppressing potential noise, ensuring reliable detection performance for the majority of datasets without prior knowledge.

D. Ablation Experiment

We follow the ablation study in [60] to demonstrate the effectiveness of our proposed multi-kernel information fusion model by comparing experimental results where the multi-kernelized



(a) Datasets from Breast to Tic2



(b) Datasets from Tic3 to Sick

Fig. 3. AUC performance curve of MKFGOD with varying hyperparameter k .

fuzzy relation in MKFGOD is replaced with a single-kernelized fuzzy relation.

As illustrated in Fig. 4 and Table IV, MKFGOD consistently outperforms the single-kernel method across all 24 datasets. This enhancement in detection robustness stems from the “multi-view” characterization provided by the fusion mechanism. A

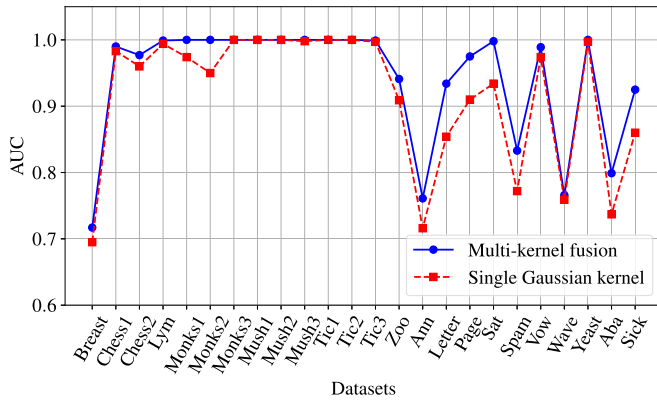


Fig. 4. Ablation study of the multi-kernel information fusion model in MKF-GOD.

TABLE IV

AUC RESULTS COMPARISON OF MKFGOD USING DIFFERENT FUSION STRATEGIES AND SINGLE-KERNEL METHOD (GAUSSIAN KERNEL) ACROSS 24 DATASETS. THE BEST AUC RESULT IN EACH ROW IS **BOLD**ED.

Datasets	Maximum	Minimum	Mean	Multiply	Weighted	Single-kernel
Breast	0.715	0.717	0.717	0.717	0.717	0.695
Chess1	0.990	0.989	0.989	0.989	0.989	0.983
Chess2	0.977	0.976	0.976	0.976	0.976	0.960
Lym	0.999	0.998	0.998	0.999	0.998	0.994
Monks1	1.000	1.000	1.000	1.000	1.000	0.974
Monks2	1.000	1.000	1.000	1.000	1.000	0.950
Monks3	1.000	1.000	1.000	1.000	1.000	1.000
Mush1	1.000	1.000	1.000	1.000	1.000	1.000
Mush2	1.000	1.000	1.000	1.000	1.000	1.000
Mush3	1.000	1.000	1.000	1.000	1.000	0.998
Tic1	1.000	1.000	1.000	1.000	1.000	1.000
Tic2	1.000	1.000	1.000	1.000	1.000	1.000
Tic3	0.999	0.999	0.999	0.999	0.999	0.997
Zoo	0.941	0.926	0.926	0.926	0.925	0.909
Ann	0.760	0.761	0.761	0.761	0.761	0.716
Letter	0.931	0.934	0.933	0.933	0.933	0.854
Page	0.975	0.974	0.975	0.974	0.974	0.910
Sat	0.998	0.997	0.998	0.998	0.998	0.934
Spam	0.830	0.833	0.832	0.832	0.833	0.772
Vow	0.988	0.989	0.989	0.989	0.989	0.974
Wave	0.766	0.765	0.766	0.765	0.766	0.759
Yeast	1.000	1.000	1.000	1.000	1.000	0.997
Aba	0.799	0.792	0.794	0.794	0.793	0.737
Sick	0.925	0.920	0.922	0.916	0.921	0.860
Average	0.941	0.940	0.941	0.940	0.940	0.916

single kernel (e.g., Gaussian) may fail to capture the underlying manifold of data with complex decay rates or multi-scale distributions, leading to “kernel failure” in certain scenarios. By integrating Gaussian, Exponential, and Rational quadratic kernels, MKFGOD achieves an average AUC of 0.942, a significant improvement of 2.6% over the single-kernel baseline (0.916). The minimal performance variance among different fusion strategies further confirms that the multi-kernel framework provides a more reliable and stable foundation for density estimation, ensuring robust performance even without prior knowledge of the data distribution.

In MKFGOD, we develop a novel multi-kernel information fusion model that integrates multi-kernel information using five distinct fusion strategies, enabling the construction of diverse multi-kernelized fuzzy relations. To investigate the impact of fusion strategies including maximum, minimum, mean, multiply, and weighted fusions on MKFGOD’s performance, we compare the AUC results of MKFGOD across all datasets using different fusion strategies. The comparative results, presented in Table IV,

indicate that regardless of the fusion strategy applied, all significantly outperform the single-kernel method, confirming the substantial improvement in detection performance attributable to multi-kernel information fusion. Furthermore, the minimal performance differences among the five fusion strategies suggest that the proposed fusion model is robust with respect to the choice of fusion strategy. For practical applications where prior knowledge of the data distribution is unavailable, we recommend using **Mean fusion** as the default strategy. It provides a balanced and smoothed estimation that mitigates individual kernel bias with high efficiency. Alternatively, **Weighted fusion** can be prioritized for complex datasets as it adaptively leverages fuzzy information entropy to emphasize kernels with stronger discriminative capabilities.

VI. CONCLUSION

In this study, we explore enhancing existing kernelized fuzzy outlier detectors by integrating FIG theory with multi-kernel information fusion. Specifically, we propose the Multi-Kernelized Fuzzy Granular Outlier Detector (MKFGOD), which establishes a novel multi-kernel information fusion model and incorporates various fusion strategies to derive multi-kernelized fuzzy relations. Ultimately, outliers are detected by evaluating the deviation between a sample’s multi-kernelized fuzzy neighborhood density and that of its multi-kernelized FkNN.

Although MKFGOD demonstrates desirable performance, it has certain limitations that merit further investigation. First, the parameters of the kernel functions are currently fixed, and the hyperparameter k requires manual tuning for different datasets to achieve optimal performance, which constrains the model’s adaptability and practical utility. Second, regarding computational complexity, the construction of the fuzzy relation matrix operates in $O(n^2)$, which may become a bottleneck for massive datasets. Third, the method relies on the assumption that all input attributes are relevant. In high-dimensional spaces with significant noise or irrelevant features, the discriminative power of the kernelized relations may be compromised. In future work, we aim to address these challenges through three main directions: 1) developing a multi-kernel fusion model capable of learning parameters adaptively by exploring optimization techniques such as genetic algorithms or reinforcement learning to automatically tune critical parameters (e.g., neighborhood size k), thereby reducing the dependency on manual tuning; 2) incorporating specific acceleration strategies for large-scale data, such as using GBC to reduce the sample cardinality n and employing kernel approximation techniques (e.g., the Nyström method) to achieve near-linear complexity; and 3) integrating feature weighting or sparse feature selection mechanisms into the fusion process to enhance robustness against noisy or irrelevant attributes.

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